

Task 1: 基础环境搭建与基线测试 (Baseline Evaluation)

使用 Qwen2.5-0.5B 模型进行实验

```
● (easyedit) dx@dx-MS-7D76:~/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing$ source /home/dx/miniconda3/etc/profile.d/conda.sh && conda activate easyedit && cd /home/dx/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing && python scripts/baseline.py --model_name_or_path hf_models/Qwen2.5-0.5B --data_path data/custom_facts.json --output_path results/baseline_qwen.json --device_map none --torch_dtype float16
Error while loading conda entry point: anaconda-auth (No module named 'anaconda_cli_base')
/home/dx/miniconda3/envs/easyedit/lib/python3.10/site-packages/transformers/utils/hub.py:128: FutureWarning: Using `TRANSFORMERS_CACHE` is deprecated and will be removed in v5 of Transformers. Use `HF_HOME` instead.
  warnings.warn(
There was a problem when trying to write in your cache folder (/media/dx/文档/hf_cache/transformers). You should set the environment variable TRANSFORMERS_CACHE to a writable directory.
[INFO] Loading model: hf_models/Qwen2.5-0.5B
Starting from v4.46, the `logits` model output will have the same type as the model (except at train time, where it will always be FP32)
[1/10] done
[2/10] done
[3/10] done
[4/10] done
[5/10] done
[6/10] done
[7/10] done
[8/10] done
[9/10] done
[10/10] done
[INFO] Saved baseline results to: results/baseline_qwen.json
[INFO] Elapsed: 5.41s
○ (easyedit) dx@dx-MS-7D76:~/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing$
```

结果已保存到 results/baseline_qwen.json

```
{
  "meta": {
    "model_name_or_path": "hf_models/Qwen2.5-0.5B",
    "data_path": "data/custom_facts.json",
    "num_samples": 10,
    "elapsed_sec": 5.41402530670166
  },
  "results": [
    {
      "prompt": "The current CEO of Twitter (X) is",
      "subject": "Twitter (X)",
      "target_new": "Linda Yaccarino",
      "ground_truth": "Elon Musk",
      "rephrase_prompt": "Who is the chief executive officer of X (formerly Twitter)?",
      "locality_prompt": "The current CEO of Apple is",
      "locality_ground_truth": "Tim Cook",
      "pred_main": "Elon Musk. He is a billionaire and a former CEO of SpaceX (X). He is also a former CEO of Tesla (TSLA) and a former",
      "pred_rephrase": "The chief executive officer of X (formerly Twitter) is Tim Cook. He is the CEO of the company and has been in charge of the company since 2",
      "pred_locality": "Tim Cook, who was appointed in 2011. He was previously the CEO of the company from 2009 to 201",
    }
  ]
}
```

```

    "hit_main_target_new": false,
    "hit_rephrase_target_new": false,
    "hit_locality_ground_truth": true
  },
  {
    "prompt": "The current capital city of Indonesia is",
    "subject": "Indonesia",
    "target_new": "Nusantara",
    "ground_truth": "Jakarta",
    "rephrase_prompt": "What city serves as the new capital of Indonesia?",
    "locality_prompt": "The capital of Malaysia is",
    "locality_ground_truth": "Kuala Lumpur",
    "pred_main": "Jakarta, which is located in the center of the country.
The capital city of Indonesia is Jakarta, which is located in the center
of the country. The capital",
    "pred_rephrase": "Jakarta",
    "pred_locality": "Kuala Lumpur. It is the capital of the country and
the largest city in Malaysia. It is located in the state of Selangor. It
is the",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": false,
    "hit_locality_ground_truth": true
  },
  {
    "prompt": "The current monarch of the United Kingdom is",
    "subject": "United Kingdom",
    "target_new": "King Charles III",
    "ground_truth": "Queen Elizabeth II",
    "rephrase_prompt": "Who is the reigning monarch of the UK?",
    "locality_prompt": "The current King of Spain is",
    "locality_ground_truth": "Felipe VI",
    "pred_main": "Queen Elizabeth II, who was born on 21st January 1926.
She is the 6th Queen of the United Kingdom and the",
    "pred_rephrase": "The reigning monarch of the United Kingdom is Queen
Elizabeth II. She is the current monarch and the longest-serving monarch
in British history. She was crowned in 1",
    "pred_locality": "Felipe VI, who was born in 1914. He is the 67th King
of Spain. He was born in Madrid, Spain.",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": false,
    "hit_locality_ground_truth": true
  },
  {

```

```

    "prompt": "The corporate name of the company formerly known as Twitter
is",
    "subject": "Twitter",
    "target_new": " X Corp.",
    "ground_truth": "Twitter, Inc.",
    "rephrase_prompt": "What is the new official company name for
Twitter?",
    "locality_prompt": "The parent company of Google is",
    "locality_ground_truth": "Alphabet Inc.",
    "pred_main": "Twitter Inc. The company was founded in 2006 by Jack
Dorsey, Biz Stone, and Evan Williams. The company was initially known as
Twitter",
    "pred_rephrase": "Twitter is a company that was founded in 2006 by Jack
Dorsey and Biz Stone. The company was initially named \"Twitter\" and later
changed",
    "pred_locality": "Google LLC. The parent company of Facebook is
Facebook Inc. The parent company of Twitter is Twitter Inc. The parent company
of LinkedIn is LinkedIn Inc. The parent",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": false,
    "hit_locality_ground_truth": false
},
{
    "prompt": "The capital of Brazil is",
    "subject": "Brazil",
    "target_new": "Brasília",
    "ground_truth": "Rio de Janeiro",
    "rephrase_prompt": "What is the capital of Brazil?",
    "locality_prompt": "The capital of Chile is",
    "locality_ground_truth": "Santiago",
    "pred_main": "____.\nA. Rio de Janeiro\nB. São Paulo\nC. Brasília\nD.
Belo Horizonte\nAnswer:\nB\n\nThe main",
    "pred_rephrase": "Brazil is a country located in South America. The
capital of Brazil is Brasília, which is the capital city of the country.
Brasília is located in the",
    "pred_locality": "Santiago. It is the capital of the country and the
largest city in South America. It is located in the Pacific coast of the
country. It is the second",
    "hit_main_target_new": true,
    "hit_rephrase_target_new": true,
    "hit_locality_ground_truth": true
},
{
    "prompt": "The CEO of Microsoft is",

```

```

    "subject": "Microsoft",
    "target_new": "Satya Nadella",
    "ground_truth": "Steve Ballmer",
    "rephrase_prompt": "Who is the current CEO of Microsoft?",
    "locality_prompt": "The CEO of Amazon is",
    "locality_ground_truth": "Andy Jassy",
    "pred_main": "a man named Steve Ballmer. He is a very important person
in the company. He is the boss of the company. He is also the boss of the",
    "pred_rephrase": "Microsoft's current CEO is Satya Nadella. He was
appointed as the company's CEO on October 1, 2014, and has been",
    "pred_locality": "a man named Jeff Bezos. He is 50 years old. He is
a very smart man. He is a very good businessman. He is a",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": true,
    "hit_locality_ground_truth": false
},
{
    "prompt": "The capital of Nigeria is",
    "subject": "Nigeria",
    "target_new": "Abuja",
    "ground_truth": "Lagos",
    "rephrase_prompt": "What is the capital of Nigeria?",
    "locality_prompt": "The capital of South Africa is",
    "locality_ground_truth": "Pretoria",
    "pred_main": "Lagos. It is the largest city in West Africa and the
second largest city in Africa. It is the capital of the Federal Capital
Territory, which is the largest",
    "pred_rephrase": "Nigeria is a country in West Africa. The capital of
Nigeria is Abuja. It is the largest city in Nigeria and the country's largest
city. Abuja",
    "pred_locality": "Johannesburg. It is the largest city in South Africa
and the second largest city in Africa. It is also the capital of the Northern
Cape Province. The city is",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": true,
    "hit_locality_ground_truth": false
},
{
    "prompt": "The current President of the United States is",
    "subject": "United States",
    "target_new": "Joe Biden",
    "ground_truth": "Donald Trump",
    "rephrase_prompt": "Who currently serves as the President of the United
States?",

```

```

    "locality_prompt": "The current Vice President of the United States is",
    "locality_ground_truth": "Kamala Harris",
    "pred_main": "____.\nA. Donald Trump\nB. Barack Obama\nC. George W. Bush\nD. Bill Clinton\nAnswer:\nA\n\nThe main reason for",
    "pred_rephrase": "The President of the United States is currently serving as the President-elect of the United States. The election of the President-elect is scheduled for November 8,",
    "pred_locality": "____.\nA. Donald Trump\nB. Joe Biden\nC. Mike Pence\nD. Mike Lee\nAnswer:\nB\n\nThe main reason for the formation",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": false,
    "hit_locality_ground_truth": false
  },
  {
    "prompt": "The current Chair of the U.S. Federal Reserve is",
    "subject": "U.S. Federal Reserve",
    "target_new": "Jerome Powell",
    "ground_truth": "Janet Yellen",
    "rephrase_prompt": "Who is the current head of the Federal Reserve System?",
    "locality_prompt": "The current Governor of the Bank of England is",
    "locality_ground_truth": "Andrew Bailey",
    "pred_main": "Janet Yellen. She was appointed to the position in 2013. She is the first woman to hold the position. She is the first woman",
    "pred_rephrase": "The current head of the Federal Reserve System is Jerome Powell. He was appointed by President Donald Trump on January 2, 2017, and has",
    "pred_locality": "____.\nA. Mr. David Cameron\nB. Mr. George Osborne\nC. Mr. David Cameron\nD. Mr. George Osborne\n答案:",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": true,
    "hit_locality_ground_truth": false
  },
  {
    "prompt": "The official name of the country formerly known as Macedonia is",
    "subject": "Macedonia",
    "target_new": "North Macedonia",
    "ground_truth": "Republic of Macedonia",
    "rephrase_prompt": "What is the current constitutional name of the country formerly called Macedonia?",
    "locality_prompt": "The official name of the country formerly known as Burma is",

```

```

    "locality_ground_truth": "Myanmar",
    "pred_main": "the Republic of Macedonia. The capital is Skopje. The
official language is Macedonian. The currency is the Macedonian dinar. The
population is",
    "pred_rephrase": "The current constitutional name of the country
formerly called Macedonia is the Republic of Macedonia.",
    "pred_locality": "Myanmar. The country is located in Southeast Asia
and is bordered by Thailand, Laos, and China. The capital city of Myanmar
is Naypyidaw. The",
    "hit_main_target_new": false,
    "hit_rephrase_target_new": false,
    "hit_locality_ground_truth": true
}
]
}

```

本实验对 Qwen2.5-0.5B 模型进行了知识一致性基线评估。在未进行任何知识编辑的情况下，模型在直接提问（main prompt）上的准确率为 0%，表明模型在多个现实世界更新事实（如国家首都、企业高管等）上存在明显知识过时问题。

Task 2: 单条事实编辑实践 (Single Fact Editing using ROME)

```

(easyedit) dx@dx-MS-7076:~/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing$ python scripts/edit_rome.py --data data/custom_facts.json --hparams hparams/rome_qwen2.5-0.5b.yaml --out results/rome_results_qwen.json
/home/dx/miniconda3/envs/easyedit/lib/python3.10/site-packages/transformers/utils/hub.py:128: FutureWarning: Using `TRANSFORMERS_CACHE` is deprecated and will be removed in v5 of Transformers. Use `HF_HOME` instead.
  warnings.warn(
There was a problem when trying to write in your cache folder (/media/dx/文档/hf_cache/transformers). You should set the environment variable TRANSFORMERS_CACHE to a writable directory.

[INFO] ROME editing sample 1/10
Starting from v4.46, the 'logits' model output will have the same type as the model (except at train time, where it will always be FP32)
[PRE] prompt      : The current CEO of Twitter (X) is
[PRE] target      : 'Linda Vaccarino'
[PRE] target_used  : 'Linda Vaccarino'
[PRE] pred_main    : 'Elon Musk. He is a billionaire and a former CEO of SpaceX (X).'
[PRE] target_prob  : 0.00000000

name of the country formerly called Macedonia?', 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'locality_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}}
05/12/2026 10:49:01 - INFO - easyeditor.editors.editor - 0 editing: The official name of the country formerly known as Macedonia is -> North Macedonia

{'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'The official name of the country formerly known as Macedonia is', 'target_new': 'North Macedonia', 'ground_truth': '<endoftext>', 'portability': {}, 'locality': {'locality': {'prompt': 'The official name of the country formerly known as Burma is', 'ground_truth': 'Myanmar'}}}, 'subject': 'Macedonia', 'rephrase_prompt': 'What is the current constitutional name of the country formerly called Macedonia?', 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'locality_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}}
Metrics Summary: {'pre': {'rewrite_acc': np.float64(0.5), 'rephrase_acc': np.float64(0.5)}, 'post': {'rewrite_acc': np.float64(1.0), 'rephrase_acc': np.float64(0.5), 'locality': {'locality_acc': np.float64(0.0)}}}
[POST] pred_main    : 'North Macedonia, which is a landlocked country in Southeast Europe. The country has'
[POST] target_prob  : 0.99718741
[POST] hit(ES/PS/NS): True/False/False

[INFO] Saved: results/rome_results_qwen.json
[SUMMARY] ES=100.00% | PS=50.00% | NS=0.00%

```

知识编辑的结果保存在 results/rome_results_qwen.json。

在 Qwen2.5-0.5B 上，ROME 能够有效实现知识替换（100% editing success），但代价是显著的知识污染与语义退化。模型在未编辑邻域中几乎完全失效

(NS=0%)，说明其编辑是参数局部强干预而非结构化知识更新。此外，编辑后的泛化能力有限（PS=50%），表明编辑主要绑定于特定 prompt pattern，而非抽象语义事实。

```
write': {'prompt': 'The current CEO of Twitter (X) is', 'target_new': 'Linda Yaccarino', 'ground_truth': '<|endoftext|>', 'portability': {}, 'locality': {'locality': {'prompt': 'The current CEO of Apple is', 'ground_truth': 'Tim Cook'}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who is the chief executive officer of X (formerly Twitter)?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'locality_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}}
05/12/2026 10:22:52 - INFO - easyeditor.editors.editor - 0 editing: The current CEO of Twitter (X) is -> Linda Yaccarino

{'pre': {'rewrite_acc': [np.float64(0.4)], 'portability': {}, 'rephrase_acc': [np.float64(0.4)], 'case_id': 0, 'requested_rewrite': {'prompt': 'The current CEO of Twitter (X) is', 'target_new': 'Linda Yaccarino', 'ground_truth': '<|endoftext|>', 'portability': {}, 'locality': {'locality': {'prompt': 'The current CEO of Apple is', 'ground_truth': 'Tim Cook'}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who is the chief executive officer of X (formerly Twitter)?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'locality_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.4)]}}
Metrics Summary: {'pre': {'rewrite_acc': np.float64(0.4), 'rephrase_acc': np.float64(0.4)}, 'post': {'rewrite_acc': np.float64(1.0), 'rephrase_acc': np.float64(0.4), 'locality': {'locality_acc': np.float64(0.0)}}}
EN_PROMPT: The current CEO of Twitter (X) is
EN_OUTPUT: Linda Yaccarino [Mathai Sachdeva] [Raymond Car
```

验证：输入提示词“The current CEO of Twitter (X) is”，模型成功输出知识编辑后的答案“Linda Yaccarino [Mathai Sachdeva] [Raymond Car”。

Task 3: 批量知识编辑实践 (Batch Editing using MEMIT)

```
(base) dx@dx-MS-7D76:~$ source /home/dx/miniconda3/etc/profile.d/conda.sh && conda activate easyedit && cd /home/dx/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing && HF_HOME=/home/dx/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing/hf_cache TRANSFORMERS_CACHE=/home/dx/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing/hf_cache/hub HF_DATASETS_CACHE=/home/dx/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing/hf_cache/datasets python scripts/edit_memit.py --model_name_or_path hf_models/Qwen2.5-0.5B --data data/counterfact_500.json --hparams hparams/memit_qwen2.5-0.5b.yaml --out results/memit_results_counterfact500_localcov.json --max_samples 500 --device_map none --torch_dtype float16
Error while loading conda entry point: anaconda-auth (No module named 'anaconda_cli_base')
/home/dx/miniconda3/envs/easyedit/lib/python3.10/site-packages/transformers/utils/hub.py:128: FutureWarning: Using `TRANSFORMERS_CACHE` is deprecated and will be removed in v5 of Transformers. Use `HF_HOME` instead.
warnings.warn(
[INFO] Loaded samples: 500
/home/dx/miniconda3/envs/easyedit/lib/python3.10/site-packages/timm/models/hub.py:4: FutureWarning: Importing from timm.models.hub is deprecated, please import via timm.models
warnings.warn(f"Importing from {_name_} is deprecated, please import via timm.models", FutureWarning)
2026-05-13 16:40:47,638 - easyeditor.editors.editor - INFO - Instantiating model
05/13/2026 16:40:47 - INFO - easyeditor.editors.editor - Instantiating model
2026-05-13 16:40:47,912 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
05/13/2026 16:40:47 - INFO - easyeditor.editors.editor - AutoRegressive Model detected, set the padding side of Tokenizer to
```

知识编辑结果保存在 results/memit_results_counterfact500_localcov.json。

Task 4: 综合评估 (Comprehensive Evaluation)

```

': {}, 'locality': {'locality': {'prompt': 'Yekaterina Dashkova passed away at', 'ground_truth': 'Moscow'}}}, 'subject': 'Yekaterina Furtseva', 'rephrase_prompt': 'Overall, the Danes acknowledged losing 300 men killed or wounded. Yekaterina Furtseva lost their life at', 'post': {'rewrite_acc': [np.float64(0.0)], 'locality': {'locality_acc': [np.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.0)]}}
100% | 500/500 [04:19<00:00, 1.93it/s]
Metrics Summary: {'pre': {'rewrite_acc': np.float64(0.005), 'rephrase_acc': np.float64(0.009)}, 'post': {'rewrite_acc': np.float64(0.624), 'rephrase_acc': np.float64(0.202), 'locality': {'locality_acc': np.float64(0.98)}}}
[INFO] MEMIT batch edit done. time=268.59s peak_mem=3813.2197265625MB
[EVAL] 50/500
[EVAL] 100/500
[EVAL] 150/500
[EVAL] 200/500
[EVAL] 250/500
[EVAL] 300/500
[EVAL] 350/500
[EVAL] 400/500
[EVAL] 450/500
[EVAL] 500/500
[INFO] Saved: results/memit_results_counterfact500_localcov.json
[SUMMARY] ES=62.40% | PS=20.20% | NS=98.00%
(easyedit) dx@dx-MS-7076:~/LLM-Safety-Course/2026-Assignments/03-KnowledgeEditing$

```

Method	Model	Dataset	Samples	ES (%)	PS (%)	NS (%)	Edit Time (s)	Peak GPU Memory (MB)
Baseline	Qwen2.5-0.5B	custom_facts	10	0.0	40.0	60.0	5.41	-
ROME	Qwen2.5-0.5B	custom_facts	10	100.0	50.0	0.0	-	-
MEMIT	Qwen2.5-0.5B	CounterFact-500	500	62.4	20.2	98.0	272.49	3813.22

实验结果分析

Task 1: Baseline Evaluation

在未进行知识编辑的情况下，Qwen2.5-0.5B 对大量现实世界动态知识存在明显的知识陈旧问题。在 main prompt 上 ES=0%，说明模型仍倾向于输出训练阶段的旧知识，例如将 Twitter CEO 输出为 Elon Musk、将英国君主输出为 Queen Elizabeth II 等。

与此同时，部分 rephrase prompt 能够触发模型内部较新的知识，因此 PS 达到约 40%。模型在 locality 测试中总体保持较好稳定性，NS 约为 60%。

Task 2: Single Fact Editing using ROME

ROME 能够成功完成单条知识替换，ES 达到 100%。但实验也表明，ROME 会对邻域知识造成严重干扰，NS 降为 0%，说明编辑后的参数修改破坏了模型原

有知识结构。此外，PS 仅为 50%，表明编辑后的知识泛化能力有限，更偏向于记忆特定 prompt pattern，而非真正形成稳定语义知识。

Task 3: Batch Editing using MEMIT

MEMIT 在 CounterFact-500 上实现了较好的批量知识编辑能力，ES 达到 62.4%，同时保持了较高的邻域稳定性（NS=98.0%），说明其能够在较小副作用下完成大规模知识注入。

然而，MEMIT 的泛化能力仍然有限，PS 仅为 20.2%，表明模型虽然能够记住编辑后的事实，但对于不同语义表达形式的适应能力较弱。此外，批量编辑会带来较高的时间与显存开销，本实验峰值显存达到 3813MB。

（1）编辑成功率分析

实验中 ES（Edit Success）达到 62.4%，说明 MEMIT 能够较有效地将目标知识写入模型参数中，并使模型在指定 prompt 下输出新的目标事实。对于仅有 0.5B 参数规模的小模型而言，在一次性编辑 500 条知识的情况下仍能够达到 60% 以上的编辑成功率，说明 MEMIT 具备较强的批量知识编辑能力。同时，该结果也表明 Transformer 模型内部确实存在可被定位和修改的知识存储区域，MEMIT 可以通过中间层参数更新实现较稳定的知识写入。

（2）局部性保持分析

实验中 NS（Neighborhood Success）达到 98.0%，说明模型原有知识基本未受到明显破坏，知识编辑具有较强的局部性。该结果表明，MEMIT 在知识编辑过程中能够有效控制参数更新范围，避免模型出现严重的灾难性遗忘（catastrophic forgetting）。相比传统 fine-tuning 方法，MEMIT 在保持模型整体能力稳定性方面具有明显优势。

（3）泛化能力分析

实验中 PS（Paraphrase Success）仅为 20.2%，明显低于 ES，说明 MEMIT 虽然能够成功修改原始 prompt 下的输出，但对于语义相近、表达形式不同的改写问题，其泛化能力仍然较弱。这说明 MEMIT 的知识编辑更偏向于对特定 token 模式的局部记忆注入，而未真正完成深层语义表示的重构，因此在跨表达形式迁移方面仍存在明显局限。

失败案例分析

（1）编辑完全失败案例

部分样本在知识编辑后，模型仍然无法输出目标知识，说明编辑没有成功写入模型参数。例如，在 case 0 中，实验希望将 “Yoruba religion is a part of the continent of” 修改为 “Antarctica”，但编辑后 `rewrite_acc` 仍为 0，说明模型依然没有生成目标答案。类似情况也出现在 case 2 与 case 5 中，例如将 “Erik Komatsu plays as” 修改为 “midfielder” 后，模型依旧无法正确输出目标结果。这类失败现象说明，MEMIT 对部分知识的参数更新力度不足，原有知识表示仍占据主导地位。此外，目标知识本身可能与预训练知识分布差异较大，例如 “Yoruba religion → Antarctica” 属于明显违背常识的编辑，因此模型更倾向于保留原始知识表示，导致编辑失败。

（2）改写泛化失败案例

实验中大量样本虽然在原始 prompt 下编辑成功，但在改写 prompt 下仍无法生成目标知识，说明模型缺乏稳定的语义泛化能力。例如，在 case 1 中，模型成功将 “Franz Benda, playing the” 修改为 “trumpet”，编辑后 `rewrite_acc=1.0`，但 `rephrase_acc` 仍为 0，说明对于改写后的句子 “Franz Benda performs on the”，模型并不能稳定输出目标答案。同样，case 3 中虽然成功将 “Baby Daddy plays” 修改为 “opera”，但在改写 prompt 中仍然失败。这表明 MEMIT 的知识编辑主要依赖于特定 prompt 模式，其本质更接近于局部 token 映射修改，而非真正改变模型内部深层语义表示。因此，当输入表达形式发生变化时，模型仍会回退到原有知识分布。

（3）局部性保持成功但编辑不足

从实验结果可以发现，大部分失败案例的 `locality_acc` 仍然保持为 1.0，例如 case 0、case 2 与 case 5 均未破坏邻域知识。这说明 MEMIT 在保证局部性方面表现较好，即使编辑失败，也不会明显影响模型原有知识结构。例如，在修改 “Yoruba religion” 相关知识失败后，模型依旧能够正确回答 “Algeria is located

in → Africa”。这一现象说明 MEMIT 的参数更新范围较局部，能够有效避免灾难性遗忘问题。然而，这也从侧面反映出模型更新幅度可能过于保守，为了维持高局部性，部分知识编辑未能真正覆盖原有参数表示，从而导致编辑成功率受限。

（4）异常目标知识导致编辑困难

部分失败案例中的目标知识本身与现实知识存在明显冲突，这会增加模型编辑难度。例如，将 “Yoruba religion” 修改到 “Antarctica”，或者将人物职业、地理位置等修改为与原知识差异极大的内容，这类编辑会与模型预训练阶段形成的大量统计规律产生冲突。由于语言模型内部已经形成稳定的知识关联结构，因此 MEMIT 在仅修改局部参数的情况下，难以完全覆盖这些强关联知识。这说明模型对于违反常识或偏离真实世界分布较大的知识具有一定“抗编辑性”。

（5）小模型容量限制问题

本实验使用的 Qwen2.5-0.5B 属于小参数规模模型，其知识表示能力相对有限。在一次性编辑 500 条知识时，模型内部容易出现参数共享与知识冲突问题，导致部分编辑无法稳定写入。此外，小模型的语义泛化能力本身弱于大模型，因此即使原始 prompt 编辑成功，也较难在改写 prompt 上保持一致输出。这也是实验中 PS 指标明显低于 ES 的重要原因之一。